Small Ruminant Research 260 (2026) 107790

Contents lists available at ScienceDirect

Small Ruminant Research

journal homepage: www.elsevier.com/locate/smallrumres

Replacing corn silage with cactus cladodes, sugarcane bagasse, and urea reduces nitrogen efficiency and microbial protein synthesis in sheep

Caio Cesar Carneiro dos Santos [a,*], Kaique Renan da Silva Salvador [a], Ana Clara Silva Pinheiro Leite [a], Melissa Silva Paiva [a], Michelle Christina Bernardo de Siqueira [a], Antônio Jorge Corrêa de Araújo [a], Gilmar Natanael de Souza [a], José Augusto Bastos Afonso da Silva [b], Francisco Fernando Ramos de Carvalho [a], João Paulo Ismério dos Santos Monnerat [a], Marcelo de Andrade Ferreira [a]

[a] *Department of Animal Science, Federal Rural University of Pernambuco, Rua Dom Manoel de Medeiros, Dois Irmãos, Recife, Pernambuco 52171-900, Brazil*

[b] *Garanhuns Cattle Clinic, Academic Unit of Garanhuns, Federal Rural University of Pernambuco, Av. Bom Pastor, Boa Vista, Garanhuns, Pernambuco 55292-270, Brazil*

ARTICLE INFO

Keywords:
Cactaceae
Neutral detergent fiber
Non-protein nitrogen
Rumen fermentation
Semi-arid conditions

ABSTRACT

The objective of this study was to evaluate the effect of replacing corn silage with cactus cladodes (*Opuntia stricta* (Haw.) Haw) plus sugarcane bagasse and urea in diets with similar neutral detergent fiber (NDF) contents on potentially digestible dry matter intake (pdDM), ruminal fermentation, nitrogen utilization efficiency, and microbial protein synthesis in sheep. Five substitution levels (0, 175, 350, 525, and 700 g/kg DM) were tested as experimental treatments, using five rumen-fistulated sheep (average body weight of 37.5 ± 5.57 kg) distributed in a 5 × 5 Latin square design. The intake of pdDM decreased linearly with increasing levels of substitution ($P < 0.01$). Ruminal pH was not affected by the replacement (6.2 ± 0.04). Ruminal ammonia nitrogen (NH_3-N) increased linearly with increasing substitution levels (15.10–19.08 mg/dL; $P = 0.034$). Additionally, a significant interaction between substitution levels and collection time was observed, with higher NH_3-N peaks as the replacement level increased (19.00–27.12 mg/dL; $P < 0.001$). Conversely, nitrogen intake ($P = 0.048$), fecal nitrogen ($P = 0.022$), retained nitrogen (g/day, $P = 0.039$; % of intake, $P = 0.034$), and microbial protein synthesis ($P = 0.006$) decreased with increasing substitution levels. In contrast, urinary nitrogen as a percentage of intake and blood urea nitrogen increased linearly. Under the experimental conditions evaluated in this study, replacing corn silage with the tested integrated roughage strategy formulated solely based on total NDF concentration reduced fermentable energy availability, nitrogen use efficiency, and microbial protein synthesis.

1. Introduction

The cactus cladodes (*Opuntia* spp.) exhibits high drought tolerance and a chemical composition characterized by elevated water content and a high concentration of non-fibrous carbohydrates (NFC) (Dutra et al., 2024). In this context, this cactus has been widely cultivated and incorporated into ruminant diets in semi-arid regions — especially during periods of low forage availability, both in quantity and quality — with the aim of partially or completely replacing traditional roughages (as silages, hays, and fresh forages) (Munhamed et al. 2021; Sánchez et al., 2022; Lima et al., 2023).

However, despite its well-recognized potential as an energy and water source, cactus cladodes have a low neutral detergent fiber (NDF) content, a characteristic that may compromise chewing activity, ruminal motility, and the stability of the fermentative environment when used as the sole roughage in diet (Siqueira et al., 2022). Therefore, its association with sources of structural fiber is essential to ensure proper rumination stimulation, maintenance of ruminal pH, and optimal rumen

* Corresponding author.
E-mail addresses: caio.csantos.94@gmail.com (C.C.C. dos Santos), kaiquersalvador@outlook.com (K.R. da Silva Salvador), anaclarapinheiroleite@gmai.com (A.C.S.P. Leite), mellissa.paiva@ufrpe.br (M.S. Paiva), michelle.siqueira2@gmail.com (M.C.B. de Siqueira), tonico.jorge04@gmail.com (A.J.C. de Araújo), gilmar.natanael05@gmail.com (G.N. de Souza), jose.basilva@ufrpe.br (J.A.B.A. da Silva), francisco.rcarvalho@ufrpe.br (F.F.R. de Carvalho), joao.monnerat@dz.ufrpe.br (J.P.I. dos Santos Monnerat), marcelo.aferreira@ufrpe.br (M. de Andrade Ferreira).

https://doi.org/10.1016/j.smallrumres.2026.107790
Received 29 December 2025; Received in revised form 27 March 2026; Accepted 21 April 2026
Available online 24 April 2026
0921-4488/

function (Pastorelli et al., 2022; Dutra et al., 2024).

In this context, in different parts of the world — particularly in arid and semiarid zones of Africa, Asia, and Latin America — the association of cactus cladodes with fibrous ingredients of low degradability, such as wheat straw, barley straw, oat straw, and sugarcane bagasse, has been adopted as a nutritional strategy (Ben Salem et al., 2004; De Waal et al. 2015; Pastorelli et al., 2022; Gebeyew et al., 2025). This combination aims to correct the deficit of physically effective fiber in cactus pear, allowing its inclusion at higher proportions in the diet while also promoting the use of locally available agricultural by-products.

For example, the use of the cactus cladodes 'Orelha de elefante mexicana' (OEM) (*Opuntia stricta* (Haw.) Haw) in association with sugarcane bagasse and urea, as potential substitutes for different types of roughage, has shown promising results, particularly by increasing dry matter (DM), organic matter (OM), and NFC intake and digestibility (Campelo-Lima et al., 2022; Siqueira et al., 2022; Medeiros et al., 2024). Although previous studies have demonstrated promising results when OEM cactus is associated with sugarcane bagasse and urea as a roughage alternative, these comparisons are frequently conducted among diets with inherently different NDF contents, which may confound the interpretation of intake, fermentation, and nitrogen utilization responses (Siqueira et al., 2019; Siqueira et al., 2022; Medeiros et al., 2024).

As NDF plays a central role in regulating voluntary dry matter intake and ruminal dynamics — directly influencing ruminal pH, ruminal ammonia availability, microbial protein synthesis, and the efficiency of dietary nitrogen utilization — differences in total NDF levels may mask the true nutritional consequences of replacing traditional forages with alternative fibrous strategies (Rufino et al., 2016; Shi et al., 2023). Therefore, evaluating this substitution under conditions of similar total NDF content becomes essential to reduce biases associated with total fiber concentration and to allow proper interpretation of the effects arising from changes in the degradable fraction of NDF.

In this context, we hypothesized that the progressive replacement of corn silage by an integrated roughage strategy composed of OEM cactus, sugarcane bagasse, and urea—formulated to maintain similar total NDF levels across diets—would not impair ruminal parameters in sheep. Thus, the objective of this study was to evaluate the nutritional and ruminal consequences of this replacement strategy on potentially digestible dry matter intake, ruminal fermentation, nitrogen utilization efficiency, and microbial protein synthesis.

2. Materials and methods

The experiment was conducted at the Federal Rural University of Pernambuco (UFRPE), in the Confinement Facility of the Small Ruminant Production Unit, and the analyses of feed, orts, and fecal samples were performed in the Animal Nutrition Laboratory of the Department of Animal Science (DZ).The management and handling of the animals followed the guidelines of the Institutional Animal Care and Use Committee (IACUC) of UFRPE (IACUC no. 8811290322; ID 001020).

2.1. Animals, management and experimental design

Five Santa Ines sheep fitted with ruminal fistulas were used, with an average body weight of 37.5 ± 5.57 kg (SD). After weighing, identification and antiparasitic treatment (for both ecto- and endoparasites), the animals were housed in individual pens measuring 1.0 m × 1.8 m, each equipped with a feed bunk and a drinking trough.

The animals were assigned to a 5 × 5 Latin square design, consisting of five animals, five treatments, and five experimental periods. Each experimental period lasted 22 days, with 14 days for dietary adaptation and 8 days for data and sample collection, totaling 110 experimental days. Prior to the start of the trial, a 10-day adaptation period to housing and handling conditions was carried out.

The diets were offered to the animals *ad libitum*, allowing for refusals of 5–10% of DM. Feed was provided twice daily, at 08:00 and 16:00 h, and clean, fresh water was always available to the animals. Feed and orts were weighed daily to adjust feed allocation based on the previous daily DM intake, throughout all experimental periods.

2.2. Ingredients and formulation of diets

The chemical composition of the ingredients used in diet formulation, as well as the ingredient proportions and chemical composition of the experimental diets, are presented in Tables 1 and 2, respectively. The experimental diets consisted of different levels of replacement of corn silage with Orelha de elefante mexicana forage cactus (*Opuntia stricta* (Haw.) Haw) associated with sugarcane bagasse and urea, at replacement levels of 0, 175, 350, 525, and 700 g/kg of DM, in addition to the concentrate ingredients (ground corn and soybean meal), common salt, and a commercial mineral premix (Ovinofós®, Tortuga/DSM, São Paulo, Brazil). All diets were formulated to maintain a roughage-to-concentrate ratio of 70:30. The diets were formulated based on the NDF content of the diet with 0 g/kg DM replacement (corn silage as the sole roughage), in order to maintain similar NDF levels across all experimental diets (0, 175, 350, 525, and 700 g/kg DM), while respecting the proportional relationship between OEM cactus and sugarcane bagasse (44:56, respectively). The experimental unit was defined as the dietary replacement strategy level, and not as individual ingredient effects. Therefore, the statistical model evaluates the response to increasing substitution levels rather than the isolated contribution of OEM, urea, or sugarcane bagasse.

2.3. Sample and data collection

Feed and orts were collected from the 15th to the 22nd day of each experimental period. Daily samples were stored in a freezer (-4°C) for subsequent chemical analyses in the laboratory. Specifically, on the 16th day of each experimental period, ruminal fluid was collected before feeding (0 h) and at 2, 4, and 6 h after the morning meal. Ruminal pH was measured immediately after sampling using a digital potentiometer (Kasavi®, Model K39–0014P, Taiwan). Ruminal fluid was acidified with 2 mL of hydrochloric acid (6 N), and subsamples (40 mL) were frozen at −20°C for later determination of ruminal ammonia nitrogen (NH_3-N) concentration. The determination of NH_3-N was performed according to the method described by Fenner (1965) and adapted by Detmann et al. (2021), and the resulting values were corrected using the equation proposed by Souza et al. (2013).

From the 17th to the 19th day of each experimental period, total fecal collection was performed to estimate digestible DM (DDM) and

Table 1
Chemical composition of the ingredients used in experimental diets.

Items	Corn Silage	Cactus Cladodes (OEM)	Sugarcane Bagasse	Soybean Meal	Ground Corn
		Chemical composition., g/kg DM			
Dry Matter[a]	288.3	106.8	539.7	878.4	864.4
Organic Matter	941.2	891.1	942.8	926.7	964.8
Ashes	58.8	108.0	57.2	73.3	15.1
Crude Protein	61.0	55.0	18.7	493.4	102.7
NDFap	591.3	252.3	850.5	146.0	130.0
pdNDF	422.5	139.5	403.1	128.7	113.9
iNDF	168.8	112.8	447.4	17.3	16.1
NFC	257.9	566.3	62.0	268.2	712.1

[a] g/kg of fresh matter; OEM = "Orelha de Elefante mexicana" *(Opuntia stricta (Haw.) Haw)*; NDFap = Neutral detergent fiber corrected for ash and protein; pdNDF = Potentially degradable neutral detergent fiber; iNDF = Indigestible neutral detergent fiber; NFC = Non-fibrous carbohydrates.

Table 2
Ingredient proportions and chemical composition of experimental diets.

Items	*Replacement Levels (g/kg DM)*				
	0	175	350	525	700
Ingredients, g/kg DM					
Corn silage	697.5	523.0	348.6	174.2	0.0
OEM	0.0	74.8	149.5	224.2	299.0
Sugarcane bagasse	0.0	97.7	195.4	293.1	390.5
Ground corn	132.0	132.0	132.0	132.0	132.0
Soybean meal	150.0	150.0	150.0	150.0	150.0
Urea/Ammonium sulfate[a]	0.0	2.0	4.0	6.0	8.0
Common salt	5.1	5.1	5.1	5.1	5.1
Mineral supplement[b]	15.4	15.4	15.4	15.4	15.4
Chemical composition, g/kg DM					
Dry matter[c]	363.0	332.0	299.2	274.6	253.4
Organic matter	925.5	922.0	918.5	915.1	911.6
Crude protein	130.1	130.7	131.3	131.9	132.5
NDFap	451.5	450.3	449.1	447.9	446.6
pdNDF	329.1	305.2	281.3	257.4	233.4
iNDF	122.4	145.1	167.8	190.5	213.1
NFC	316.3	321.1	325.7	330.4	335.2

[a] = Proportion of urea to ammonium sulfate (AS): 9 parts urea to 1 part ammonium sulfate;
[b] = Chemical composition of the mineral supplement: Ca 120 g/kg, P – 87 g/kg, S – 18 g/kg, Mg – 1300 mg/kg, Na – 147 g/kg, Co – 40 mg/kg, Cu – 590 mg/kg, I – 80 mg/kg, Se – 15 mg/kg, Zn – 3800 mg/kg, F (max.) – 870 mg/kg;
[c] = g/kg of natural matter; OEM cactus = Orelha de elefante mexicana cactus cladodes *(Opuntia stricta* (Haw.) Haw); NDFap = Neutral detergent fiber corrected for ash and protein; $_{pd}$NDF = Potentially degradable neutral detergent fiber; iNDF = Indigestible neutral detergent fiber; NFC = non-fibrous carbohydrates.

TDN and fecal nitrogen (fecal N) excretion. Feces were collected using fecal collection bags, weighed, and quantified over a 24-hour period. From the total fecal output of each day, a subsample of approximately 20% was retained for further analysis.

During the same period (17th to 19th day), total urine collection was also carried out over 24 h, using collecting funnels attached to the animals' genital region to direct the urine into a plastic container. The pH of the urine was measured every 6 h and maintained below 3 to prevent bacterial degradation of urinary purine bases and the precipitation of uric acid. At the end of each 24-hour collection period over the three collection days, the total urine volume was measured, and a composite sample was prepared from these daily collections. An aliquot equivalent to 10% of the total volume from each period was obtained and stored at −20°C for subsequent analyses of uric acid, allantoin, and nitrogen balance.

On the 21st day of each experimental period, blood samples were collected from the animals four hours after the morning feeding, via jugular vein puncture, using Vacutainer® tubes containing anticoagulant (heparin). Samples were immediately centrifuged (2500 rpm for 10 min) to obtain serum for the determination of blood urea concentration (mg/dL), using a LAB Max 250® analyzer and LAB Test® commercial kits.

2.4. Chemical analysis and experimental calculations

The feed, orts, and fecal samples obtained during the collection period were thawed and pre-dried in a forced-air oven at 55°C for 72 h or until reaching constant weight. A composite sample was then prepared for each animal, treatment, and experimental period. These composite samples were then processed using a knife mill (Marconi®, model MA 340, Brazil), employing sieves with 1 mm and 2 mm pore sizes. The 2 mm pore sieve was used exclusively for estimating iNDF, whereas the 1-mm sieve was employed for all other chemical analyses.

The determinations of DM (AOAC 934.01), ash (AOAC 942.05), crude protein (CP; AOAC 954.01), and ether extract (EE; AOAC 920.39) were analyzed according to the procedures described by AOAC (2023). NDF was determined according to Van Soest et al. (1991), without the use of sodium sulfite, and using thermostable α-amylase, as recommended by Mertens (2002). The NDF residue was corrected for ash (NDFa) and for neutral detergent insoluble protein (NDIP), according to the methods described by Mertens (2002) and Licitra et al. (1996), respectively. Subsequently, NDF corrected for ash and protein (NDFap) was calculated using the equation proposed by Detmann and Valadares Filho (2010): NDFap (g/kg MS) = NDF – (NDFa + NDIP).

For the iNDF estimation, the samples previously oven-dried and ground through a 2-mm sieve were placed into nonwoven fabric bags (TNT, 100 g/m^2), maintaining a sample-to-surface ratio of up to 20 mg DM/cm^2. The bags were incubated in the rumen of a fistulated bovine for 288 h. Nylon mesh bags were used to hold the sample bags inside the rumen, allowing their complete recovery at the end of the incubation period. After incubation, the bags were rinsed in running water until the soluble residues were completely removed. After that, they oven-dried at 55–60 °C to constant weight and subjected to neutral detergent extraction according to the procedure of Van Soest et al. (1991) (without the use of sodium sulfite). The iNDF value was expressed as the percentage of the residue remaining in relation to the initial DM, representing the fraction of NDF that resists microbial degradation in the rumen.

The rumen-incubating bovine was fed a maintenance diet consisting of approximately 70% forage and 30% concentrate (DM basis), offered twice daily to ensure stable ruminal conditions and adequate fibrolytic microbial activity throughout the 288 h incubation period. The potentially digestible neutral detergent fiber (pdNDF) of each dietary component was estimated using the following equation: pdNDF=NDFap−iNDF.

For the estimation of NFC in the diets containing urea (175, 350, 525, and 700 g/kg DM replacement levels), the equation proposed by Mertens (1997), adapted by Hall (2000) and described by Detmann et al. (2021), was used: NFC(%/DM) = 100 −ASH−EE−NDFap−(CP−CPU+U). Where CPU = crude protein derived from urea and U = amount of urea. For the estimation of NFC in the diet without urea (0% replacement), the equation proposed by Mertens (1997), with modifications by Detmann and Valadares Filho (2010), was applied: NFC(%/DM) = 100 −ASH−EE−NDFap−CP. To estimate TDN, the adapted equation described by Weiss et al. (1992) was used: TDN(g/kg) =CPd +NDFapd +NFCd + (EEd ×2.25). Where the d refers to the digestible fraction of each nutrient.

Allantoin concentration was determined according to Chen and Gomes (1992), using a colorimetric method with absorbance read at 522 nm in a UV–visible spectrophotometer. Uric acid concentration was measured using a LAB Max 250® analyzer and a specific commercial kit (LAB Test®) for uric acid determination in urine samples. Xanthine and hypoxanthine were estimated as 7.5% of the total purine derivatives (TPD) excreted, based on the expected proportional contribution of these metabolites in sheep and goats (5–10% of total TPD) (Chen and Gomes, 1992; BR-Caprinos & Ovinos, 2024).

The absorption of microbial purines (mmol/day) and the intestinal flow of microbial nitrogen compounds (g/day) were calculated using the mathematical model proposed by Chen and Gomes (1992). Total urinary excretion of purine derivatives (TPD) was obtained as the sum of allantoin, uric acid, and the estimated xanthine + hypoxanthine values. Microbial protein synthesis (MPS; g/day) was determined by multiplying the absorbed microbial purines by the constant 6.25. The efficiency of microbial protein synthesis was calculated by dividing the total MPS by the TDN intake, expressed as MPS (g MPS /kg TDN).

Total nitrogen in feed ingredients, orts, feces, and urine was determined according to the Kjeldahl method (AOAC 954.01; AOAC, 2023). For nitrogen balance, digested nitrogen (g/day) was calculated as the difference between total nitrogen intake (N intake) and fecal nitrogen excretion (fecal N). Retained nitrogen (g/day) was calculated as the difference between N intake and the sum of fecal N and urinary N. The proportions of digested and retained nitrogen relative to intake (% of intake) were calculated by expressing the respective digested and retained nitrogen values as a percentage of total nitrogen intake.

2.5. Statistical analysis

The statistical model used was: Yijk = μ + Si + Aj + Pk + εijk. Where: Yijk = response variable measured in animal j, during period k, subjected to treatment i; μ = overall mean; Si = effect of treatment i (fixed effect); Aj = effect of animal j (random effect); Pk = effect of experimental period k (random effect); and εijk = unobservable random error, assumed to be normally distributed. The statistical analyses were performed using the Statistical Analysis System software (SAS version 9.4 M8).

Analysis of variance and regression were performed, except for ruminal pH and NH_3-N, for which sampling time was considered as a repeated measure. The behavior of these variables over time was analyzed using the PROC MIXED procedure. Orthogonal polynomial contrasts were fitted to the quantitative treatment levels to assess linear and quadratic trends. The remaining data were analyzed using the PROC MIXED of SAS, including linear and quadratic terms in the model. Residuals were evaluated for normality and homogeneity of variance prior to analysis, and no violations requiring data transformation were detected. A significant level of 5% was adopted for Type I error.

Additionally, mixed model regression analyses were conducted to evaluate the quantitative relationships between dietary iNDF concentration (treated as a continuous variable) and selected response variables (potentially digestible dry matter intake (pdDM) and microbial protein synthesis). In these analyses, animal and period were included as random effects to account for the Latin square design.

3. Results

Daily intake of pdDM (1078.9–741.9 g/day, $P < 0.01$), digestible dry matter (DDM; 801.4–543.6 g/day, $P < 0.01$), digestible organic matter (DOM; 770.1–483.1 g/day, $P = _{0.012}$), digestible crude protein (DCP; 118.3–95.5 g/day, $P = 0.044$), digestible neutral detergent fiber (DNDF; 292.3–157.3 g/day, $P < 0.01$), digestible total carbohydrates (DTC; 619.3–429.5 g/day, $P = 0.011$), and TDN (804.1–554.0 g/day, $P = 0.012$) decreased linearly with increasing levels of replacement (Table 3). To further quantify this response, mixed model regression analysis revealed a significant negative association between dietary iNDF concentration and potentially digestible dry matter intake ($\beta = -3.67 \pm 0.68$; $P < 0.001$). No effect of replacement level was observed on the daily intake of digestible non-fibrous carbohydrates (DNFC) by the animals ($P = 0.196$; Table 3).

No interaction was observed between substitution levels and sampling time for ruminal pH ($P = 0.411$; Table 4 and Fig. 1). Ruminal pH was not affected by the substitution levels ($P = 0.153$) (overall mean of 6.27 ± 0.04 SD). A quadratic response was detected over sampling time ($P < 0.01$), with the minimum ruminal pH value of 6.11 occurring at 3.73 h after feeding.

The NH_3-N increased with increasing substitution levels (linear effect, $P = 0.034$), and a significant interaction between substitution level and sampling time was observed (interaction effect, $P < 0.001$; Table 4 and Fig. 2). Mean NH_3-N concentrations ranged from 15.10 to 19.08 mg/dL across substitution levels. The highest NH_3-N peak occurred at the 700 g/kg DM substitution level (27.12 mg/dL at 2.80 h post-feeding), followed by 525 (24.84 mg/dL at 2.83 h), 350 (21.65 mg/dL at 2.88 h), 175 (20.75 mg/dL at 2.91 h), and 0 (19.00 mg/dL at 3.02 h), respectively.

Nitrogen intake (21.04–26.90 g/day; $P = 0.048$), fecal N excretion (7.57–5.82 g/day; $P = 0.022$), and retained N expressed both in g/day (12.48–8.77 g/day; $P = 0.039$) and as a percentage of intake (46.39–41.71%; $P = 0.034$) decreased linearly with increasing substitution levels. Conversely, urinary N expressed as a percentage of intake (25.45–31.31%; $P = 0.034$) and blood urea nitrogen (BUN; 13.86–19.49 mg/dL; $P = 0.012$) increased linearly. Urinary N excretion expressed in g/day was not affected by substitution level ($P = 0.896$; Table 4). Microbial protein synthesis (MPS, g/day) in this study decreased linearly ($P = 0.006$; 57.22 – 43.05 g/day), while the efficiency of microbial protein synthesis (g MPS/kg TDNI) increased linearly ($P = 0.014$) with increasing substitution levels (71.96– 78.86 g MPS/kg TDNI; Table 4). A similar negative association was observed between dietary iNDF concentration and microbial protein synthesis ($\beta = -0.164 \pm 0.030$; $P < 0.001$).

4. Discussion

Although the diets had similar contents of NDFap, the replacement levels altered the fractional composition of this component. An increase was observed in the iNDF, ranging from 122.4 to 213.1 g/kg DM, accompanied by a concomitant reduction in the pdNDF, which decreased from 329.1 to 233.4 g/kg DM (Table 2). This behavior is consistent with findings reported in the literature for diets with increasing levels of sugarcane bagasse inclusion (Silva et al. 2015; Almeida et al. 2018). According to Soares et al. (2015), sugarcane bagasse is a roughage characterized by high cell wall content, low energy value, and a high iNDF content; its increasing inclusion in the diet alters the qualitative profile of the fiber fraction, increasing iNDF and reducing pdNDF.

As described by Van Soest (1994) and Mertens (1997), an increase in the iNDF reduces the ruminal degradability of fiber and prolongs rumen digesta retention time, which may lead to a physical limitation of intake and, consequently, to a reduction in the ingestion of potentially digestible nutrients. Although the diets were formulated to maintain similar total NDF levels, the progressive inclusion of sugarcane bagasse altered the degradability profile of the fiber fraction, increasing the iNDF and reducing the availability of fermentable energy, limiting the

Table 3
Average nutrient intake values in sheep.

Items	Replacement Levels (g/kg DM)						P-value	
	0	175	350	525	700	SEM	L	Q
				Intake, g/day				
pdDM	1078.9	915.3	786.2	757.6	741.9	41.2	< 0.01	0.259
DDM	801.4	650.8	622.4	545.4	543.6	34.7	< 0.01	0.534
DOM	770.1	625.5	545.4	489.3	483.1	34.1	0.012	0.406
DCP	118.3	115.4	95.6	94.3	95.5	4.4	0.044	0.737
DNDF	292.3	236.7	213.0	170.6	157.3	10.9	< 0.01	0.484
DNFC	326.9	289.6	299.3	279.8	272.1	12.6	0.196	0.780
DTC	619.3	526.0	512.4	450.5	429.5	23.3	0.011	0.638
TDN	804.1	727.4	625.9	615.9	554.0	76.6	0.012	0.688

pdDM= Potentially digestible dry matter; DDM= Digestible dry matter; DOM= Digestible organic matter; DCP= Digestible crude protein; DNDF= Digestible neutral detergent fiber; DNFC = Digestible non-fibrous carbohydrates; DTC= Digestible total carbohydrates; TDN = Total digestible nutrients; SEM = Standard error of the mean; L = Linear effect; Q = Quadratic effect.

Table 4
Rumen fluid, nitrogen balance, blood urea nitrogen (BUN), microbial protein synthesis, and efficiency in the rumen.

Parameters	*Replacement Levels (g/kg DM)*					SEM	P-value				
							Levels		Collection time		Interaction
	0	175	350	525	700		L	Q	L	Q	
pH	6.22	6.30	6.29	6.27	6.31	0.06	0.153	0.912	< 0.001	< 0.001	0.4114
NH_3-N (mg/dL)	15.10	15.69	16.56	18.22	19.08	1.09	0.034	0.682	< 0.001	< 0.001	< 0.001
N intake (g/day)	26.90	25.23	23.27	21.97	21.04	1.01	0.048	0.808	-	-	-
N feces (g/day)	7.57	6.72	6.14	6.21	5.82	0.23	0.022	0.429	-	-	-
N urine (g/day)	6.85	6.43	6.84	6.88	6.44	0.38	0.896	0.902	-	-	-
N urina[a]	25.45	25.47	29.39	31.31	30.63	0.92	0.034	0.898	-	-	-
N retained (g/day)	12.48	12.09	10.29	8.88	8.77	0.69	0.039	0.870	-	-	-
N retained[a]	46.39	47.91	44.21	40.42	41.71	1.39	0.034	0.936	-	-	-
BUN (mg/dL)	13.86	14.25	17.12	18.11	19.49	0.78	0.012	0.989	-	-	-
MPS (g/day)	57.22	53.15	45.56	44.37	43.05	1.76	0.006	0.429	-	-	-
g MPS/kg TDN	71.96	74.08	77.14	78.45	78.86	0.97	0.014	0.539	-	-	-

[a] percentage of intake. NH_3-N = Ruminal ammonia nitrogen; MPS = Ruminal microbial protein synthesis; N = Nitrogen; g MPS/kg TDN = Efficiency of ruminal microbial protein synthesis per kg Total digestible nutrients intake; SEM = Standard error of the mean; L = Linear effect; Q = Quadratic effect.

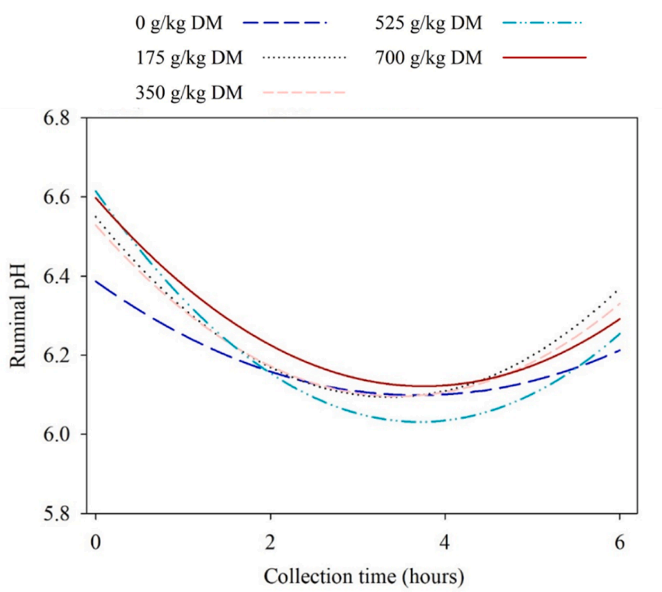

Fig. 1. Ruminal pH response over time (0–6 h after feeding) in sheep fed diets containing different replacement levels of cladodes cactus and sugarcane bagasse plus urea (0, 175, 350, 525 and 700 g/kg DM) for corn silage. Lines represent fitted regression curves for each substitution level.

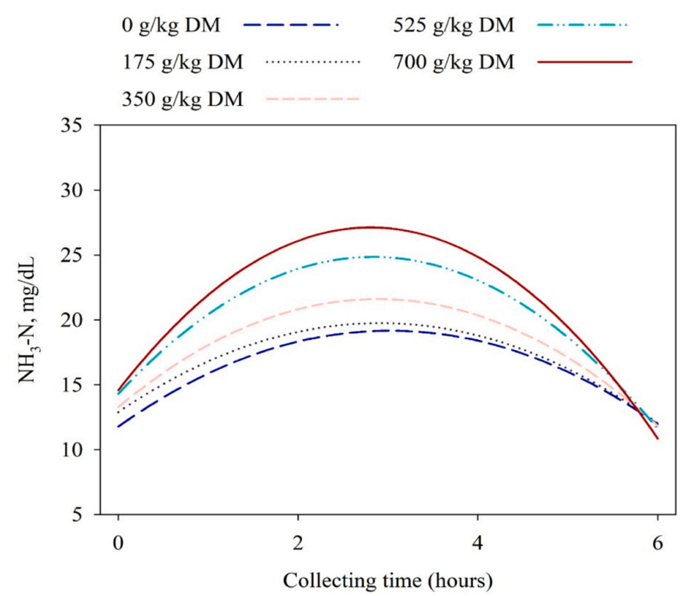

Fig. 2. Rumen ammonia nitrogen (NH_3-N, mg/dL) concentration over time (0–6 h after feeding) in sheep fed diets containing different replacement levels of cladodes cactus and sugarcane bagasse plus urea (0, 175, 350, 525 and 700 g/kg DM) for corn silage. Lines represent fitted regression curves for each substitution level.

energy intake and consumption of nutrients effectively usable by the animals (Table 3). This behavior is consistent with findings reported in literature, particularly in studies evaluating increasing levels of sugarcane bagasse inclusion or its use as the sole fiber source in high-forage diets (Freitas et al. 2018; Medeiros et al., 2024). To further quantify this response, mixed model regression analysis revealed a strong negative association between dietary iNDF concentration and potentially digestible dry matter intake ($\beta = -3.67 \pm 0.68$; $P < 0.001$), highlighting a practical limitation of formulating diets based solely on total NDF concentration without considering fiber quality.

Conversely, no significant differences were observed in the intake of digestible DNFC. This result is associated with the increase in dietary NFC concentration (316.3–335.2 g/kg DM), mainly due to the progressive inclusion of OEM cactus, an ingredient with a high NFC content (566.3 g/kg DM). In contrast, sugarcane bagasse has a low concentration of this component (62.0 g/kg DM), and the forage-to-concentrate ratio was maintained constantly across treatments (Tables 1 and 2). Moreover, NFC is a highly digestible fraction, whose utilization occurs largely independently of the digestibility of other dietary components such as NDFap. This response is consistent with previous studies evaluating the inclusion of forage cactus in ruminant diets, which report increased availability and intake of non-fibrous carbohydrates without proportional changes in fiber digestibility, due to the high soluble carbohydrate content of cactus cladodes (Santos et al. 2022; Medeiros et al. 2024).

The substitution levels did not affect the ruminal pH of the sheep (mean of 6.3 ± 0.06 SD; Table 4), which remained within the optimal range for cellulolytic bacterial activity (6.2–7.0) (Siqueira et al., 2022). This response may be partially attributed to the similar NDFap content and, possibly, the physically effective NDFap of the diets—originating mainly from corn silage and sugarcane bagasse—which likely provided a comparable stimulus for chewing and, consequently, saliva secretion.

This buffering mechanism, combined with the relatively low proportion of concentrate in the diets (approximately 30%), was likely sufficient to neutralize the acids produced during ruminal fermentation of organic matter, maintaining stable ruminal pH across substitution levels (Fig. 1) (Shi et al., 2023; Zhang et al. 2025). Although OEM inclusion increased dietary NFC, its fermentative profile resembles that of roughage feeds and did not promote ruminal acidification under the experimental conditions evaluated (Mora-Luna et al., 2022; Silva et al., 2025)

Increasing substitution levels promoted a linear increase NH_3-N concentration (15.10–19.08 mg/dL), in addition to resulting in greater post-feeding peaks during the first hours after feeding (19.00–27.12 mg/dL at 3.02 and 2.80 h, respectively) (Table 4; Fig. 2). This result indicates that, although nitrogen intake decreased with increasing substitution levels, the progressive inclusion of urea in the diets provided a greater supply of rumen-degradable nitrogen. This occurs because urea is highly soluble in the ruminal environment and is rapidly hydrolyzed by urease produced by ruminal microorganisms, releasing ammonia, which can then be incorporated into microbial cells (Ma and Faciola, 2024; Ramalho et al., 2024).

However, although the mean NH_3-N values observed in the present study fall within the range considered adequate to maximize microbial protein synthesis (15–20 mg/dL) (Niazifar et al., 2024), the post-feeding peaks exceeding this range — even when followed by subsequent declines — indicate issues related to the synchronization between the availability of fermentable energy and rumen-degradable nitrogen (Santos et al., 2022).

The linear reduction in pdDM and DOM intake (Table 3) as the replacement levels increased indicates that progressively lower amounts of fermentable energy were available in the rumen. The linear increase in iNDF content, concomitant with the reduction in the potentially degradable NDF fraction, resulted in a gradual decrease in fermentable energy availability within the rumen over time, leading to a subsequent dissociation between degradable nitrogen and available energy. This imbalance explains the increased peaks of NH_3-N observed as the replacement level increased (Fig. 3) (Zhang et al., 2020).

Accordingly, the increase in BUN, followed by the rise in urinary nitrogen expressed as a percentage of intake and the reduction in retained nitrogen observed in this study, confirms that a substantial portion of the nitrogen supplied during the NH_3-N peak was not utilized by ruminal microorganisms. Instead, it was absorbed as ammonia, converted into urea in the liver, and subsequently excreted in the urine (Hailemariam et al., 2021; Nichols et al., 2023). In addition, the reduction in fermentable energy substrate in the rumen with increasing replacement levels not only led to inefficient utilization of dietary nitrogen but also resulted in a linear decrease in microbial protein synthesis in the rumen (57.22–43.05 g/day; Table 4) (Sampaio et al., 2009).

A similar quantitative relationship was observed for microbial protein synthesis, which decreased as dietary iNDF increased ($\beta = -0.164 \pm 0.030$; $P < 0.001$). The equation proposed by Santos et al. (2021) to estimate microbial protein synthesis from TDN intake adequately predicted the values observed in the present study, reinforcing the physiological consistency of the responses detected under the replacement strategy evaluated herein (Fig. 4).

The responses observed in the present study are consistent with those reported in ruminants feed increasing levels of indigestible fiber, particularly sugarcane bagasse. Studies with lactating cows have shown that greater inclusion of this roughage reduces TDN intake and microbial protein synthesis while increasing circulating urea concentrations (Almeida et al., 2018; Freitas et al., 2018). These effects are associated with changes in fiber quality, especially the increase in iNDF and the reduction in potentially degradable fractions, which limit the availability of fermentable energy in the rumen. Consequently, the synchrony between rumen-degradable nitrogen and available energy is impaired, leading to greater ammonia accumulation and nitrogen losses, as also reported by Halako et al. (2020).

It is acknowledged that part of the increase in ruminal ammonia concentration and urinary nitrogen excretion may be attributed to the proportional inclusion of urea within the replacement strategy. However, this reflects common practical formulation scenarios in which non-protein nitrogen is used to compensate for low crude protein concentration in fibrous ingredients. The present study was not designed to separate these effects but to evaluate the integrated nutritional consequences of this replacement approach.

The results of this study indicate that replacing corn silage with OEM cactus combined with sugarcane bagasse and urea, in diets containing similar NDFap levels, does not alter ruminal pH. However, the increasing inclusion of sugarcane bagasse, used to standardize NDFap across diets, compromised the supply of fermentable energy in the rumen, resulting in reduced synchronization between fermentable energy availability and NH_3-N dynamics. This imbalance reduced the efficiency of dietary nitrogen utilization and limited microbial protein synthesis.

Considering the potential negative effects of increasing dietary iNDF on the efficiency of nitrogen compound utilization and microbial protein synthesis in the rumen, it is recommended that diet formulation consider not only the total NDF content but also the iNDF fraction and/or the pdDM content. This approach helps prevent potential reductions in

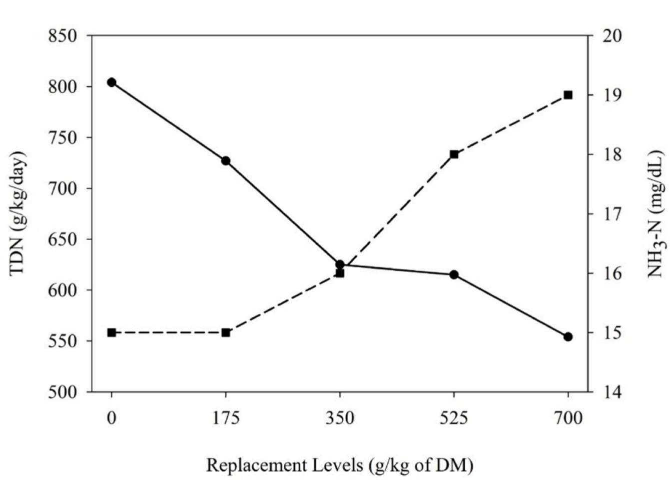

Fig. 3. Total digestible nutrients intake (TDN; g/kg/day; left y-axis) and ruminal ammonia nitrogen (NH_3-N; mg/dL; right y-axis) as a function of replacement levels (g/kg DM) of corn silage with cladodes of *Opuntia stricta* (Orelha de elefante mexicana'), sugarcane bagasse, and urea in sheep diets. Solid line with circular markers represents TDN (left axis), and dashed line with square markers represents NH_3-N (right axis).

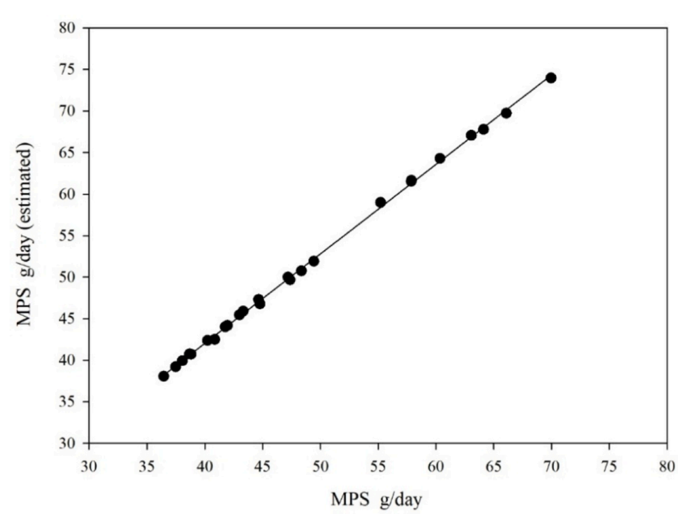

Fig. 4. Relationship between observed and estimated (using the equation developed by Santos et al. (2021)) values of microbial protein synthesis (MPS, g/day) in sheep. The solid line represents the linear regression fit, indicating a high degree of agreement between observed and predicted values (Y= 1,0773x −1,0572; R^2=0,9993).

fermentable energy availability in the rumen, particularly when sugarcane bagasse is used as the main source of NDF.

This study presents some limitations that should be considered. First, the design did not allow the separation of individual ingredient effects, as the objective was to evaluate an integrated roughage replacement strategy. Second, the short-term Latin square metabolic design using rumen-fistulated sheep did not permit evaluation of long-term performance indicators such as weight gain or feed efficiency. Therefore, future studies should investigate growth performance responses and explore formulation strategies that simultaneously consider total NDF, fiber degradability, and synchronization between fermentable energy and rumen-degradable nitrogen.

5. Conclusion

Under the experimental conditions evaluated in this study, replacing corn silage with the tested integrated roughage strategy formulated solely based on total NDF concentration reduced fermentable energy availability, nitrogen use efficiency, and microbial protein synthesis.

CRediT authorship contribution statement

Caio Cesar Carneiro dos Santos: Writing – review & editing, Investigation, Formal analysis, Data curation, Conceptualization. **Kaique Renan da Silva Salvador:** Writing – review & editing, Formal analysis. **Ana Clara Silva Pinheiro Leite:** Writing – review & editing. **Melissa Silva Paiva:** Writing – review & editing, Formal analysis. **Michelle Christina Bernardo de Siqueira:** Writing – review & editing, Formal analysis. **Antônio Jorge Corrêa de Araújo:** Writing – review & editing. **Gilmar Natanael de Souza:** Writing – review & editing. **João Paulo Ismério dos Santos Monnerat:** Writing – review & editing, Data curation, Conceptualization. **Francisco Fernando Ramos de Carvalho:** Writing – review & editing, Data curation, Conceptualization. **Marcelo de Andrade Ferreira:** Writing – review & editing, Supervision, Project administration, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was financed in part by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES) – Finance Code 001, and by the Fundação de Amparo à Ciência e Tecnologia do Estado de Pernambuco (FACEPE).

References

Almeida, G.A.P., Andrade-Ferreira, M., Lima-Silva, J., Chagas, J.C.C., Véras, A.S.C., Barros, L.J.A., Almeida, G.L.P., 2018. Sugarcane bagasse as exclusive roughage for dairy cows in smallholder livestock systems. Asian-Australas. J. Anim. Sci. 31, 379–385. https://doi.org/10.5713/ajas.17.0205.

AOAC, 2023. Official methods of analysis of the Association of Official Analytical Chemists, 22nd ed. AOAC International, Gaithersburg, MD, USA.

Ben Salem, H., Nefzaoui, A., Ben Salem, L., 2004. Spineless cactus (Opuntia ficus-indica f. inermis) and oldman saltbush (Atriplex nummularia L.) as alternative supplements for growing Barbarine lambs given straw-based diets. Small Rumin. Res. 51, 65–73. https://doi.org/10.1016/S0921-4488(03)00186-X.

BR-Caprinos & Ovinos, 2024. Exigências nutricionais de caprinos e ovinos. Ed. Sci. São Carlos Braz. https://doi.org/10.26626/9786556681849.2024B0001.

Campelo-Lima, V.H., Antas-Urbano, S., Andrade-Ferreira, M., Cavalcante-Ribeiro, P.H., Gomes-Bezerra, J.I., Cariri-Chagas, J.C., Moreira-Aguiar, E., Nascimento-Rangel, A. H., 2022. Sugarcane bagasse or elephant grass hay in diets for goats: Performance, feeding behavior and carcass characteristics. Rev. Colomb. Cienc. Pecu. 35 (1), 49–59. https://doi.org/10.17533/udea.rccp.v35n1a05.

Chen, X.B., Gomes, M.J., 1992. Estimation of microbial protein supply to sheep and cattle based on urinary excretion of purine derivatives – an overview of technical details. International Feed Resources Unit, Rowett Research Institute. Occasional Publication, Aberdeen, UK.

De Waal, H.O., Combrinck, W.J., Fouché, H.J., 2015. Preserving mashed cactus pear (Opuntia ficus-indica) fruit with wheat straw, maize hay or lucerne hay. Acta Hortic. (1067), 167–172. https://doi.org/10.17660/ActaHortic.2015.1067.22.

Detmann, E., Souza, M.A., Valadares Filho, S.C., Queiroz, A.C., Berchielli, T.T., Saliba, E. O.S., Cabral, L.S., Pina, D.S., Ladeira, M.M., Azevedo, J.A.G., 2021. Métodos para análise de alimentos. Suprema, Visconde do Rio Branco. Instituto Nacional de Ciência e Tecnologia de Ciência Animal, MG, Brazil.

Detmann, E., Valadares Filho, S.C., 2010. On the estimation of non-fibrous carbohydrates in feeds and diets. Arq. Bras. Med. Vet. Zootec. 62, 980–984. https://doi.org/10.1590/S0102-09352010000400030.

Dutra, I.C., Pires, A.J.V., Santos, B.E.F., Silva, N.V., Pio, L.P., Cruz, N.T., Carvalho-Dutra, G., 2024. Forage cactus (Opuntia ficus-indica (L.) Miller) f. Cactaceae as an alternative for ruminant feeding. Rev. Bras. Ciênc 3, 33–41. https://doi.org/10.14295/bjs.v3i9.617.

Fenner, H., 1965. Method for determining total volatile bases in rumen fluid by steam distillation. J. Dairy Sci. 48, 249–251. https://doi.org/10.3168/jds.S0022-0302(65)88206-6.

Freitas, W.R., Ferreira, M.D.A., Silva, J.L., Véras, A.S.C., Barros, L.J.A., Alves, A.M.S.V., Almeida, G.A.P., 2018. Sugarcane bagasse as only roughage for crossbred lactating cows in semiarid regions. Pesq. Agropecu. Bras. 53, 386–393. https://doi.org/10.1590/S0100-204X2018000300014.

Gebeyew, K., Mi, H., Du, R., Gao, M., Diba, D., Tang, S., He, Z., Tan, Z., 2025. Wheat straw and alfalfa hay alone or combined in a high-concentrate diet alters microbial–host interaction in the rumen of lambs. Anim. Nutr. 20, 444–457. https://doi.org/10.1016/j.aninu.2024.08.010.

Hailemariam, S.A., Zhao, W., Wang, J., Xue, M., Liu, J., 2021. Effect of synchronized energy and nitrogen supply on rumen fermentation and microbial protein synthesis. Anim. Feed Sci. Technol. 276, 114930. https://doi.org/10.1016/j.anifeedsci.2021.114930.

Halako, G.A., Yansari, A.T., Pirsaraei, Z.A., 2020. Effect of different levels of indigestible fiber and particle size on intake, rumen parameters, digestibility, chewing activity, milk yield and composition of Holstein dairy cows. J. Anim. Res. Nutr. 5, 8. https://doi.org/10.36648/2572-5459.5.1.59.

Hall, M.B., 2000. Calculation of non-structural carbohydrate content of feeds that contain non-protein nitrogen. In: Bull, 339. University of Florida, Institute of Food and Agricultural Sciences (IFAS), Gainesville, FL, USA.

Licitra, G., Hernandez, T.M., Van Soest, P.J., 1996. Standardization of procedures for nitrogen fractionation of ruminant feeds. Anim. Feed Sci. Technol. 57, 347–358. https://doi.org/10.1016/0377-8401(95)00837-3.

Lima, I.E., Monteiro, C.C., Mesquita, F.T.L., Vasconcelos, E.Q.L., Souza, M.S., Santos, D. S., Andrade-Ferreira, M., 2023. Cactus cladodes for dairy goats: what is the best fiber source? Trop. Anim. Health Prod. 55, 347. https://doi.org/10.1007/s11250-023-03771-5.

Ma, S.-W., Faciola, A.P., 2024. Impacts of slow-release urea in ruminant diets: A review. Fermentation 10, 527. https://doi.org/10.3390/fermentation10100527.

Medeiros, I.P.S., Guido, S.I., Gama, M.A.S., Silva, C.H.M., Siqueira, M.C.B., Silva, C.S., Netto, A.J., Felix, S.B., Rabelo, M.N., Santos, T.V.M., Leite, M.A.M., Ferreira, M.A., 2024. Cactus cladodes and sugarcane bagasse can partially replace earless corn silage in diets of lactating dairy cows. Dairy 5, 33–43. https://doi.org/10.3390/dairy5010003.

Mertens, D.R., 1997. Creating a system for meeting the fiber requirements of dairy cows. J. Dairy Sci. 80 (7), 1463–1481. https://doi.org/10.3168/jds.S0022-0302(97)76075-2.

Mertens, D.R., 2002. Gravimetric determination of amylase-treated neutral detergent fiber in refluxed diets in beakers or crucibles: a collaborative study. J. AOAC Int. 85 (6), 1217–1240. https://doi.org/10.1093/jaoac/85.6.1217.

Mora-Luna, R.E., Herrera-Angulo, A.M., Siqueira, M.C.B., Conceição, M.G., Chagas, J.C. C., Monteiro, C.C.F., Véras, A.S.C., Carvalho, F.F.R., Ferreira, M.A., 2022. Spineless cactus plus urea and Tifton-85 hay: maximizing the digestible organic matter intake, ruminal fermentation and nitrogen utilization of wethers in semi-arid regions. Animals 12, 401. https://doi.org/10.3390/ani12030401.

Munhamed, J.A., Batista, Â.M., Monnerat, J.P.I.S., Maciel, M.V., Lopes, L.A., Silva, T.G. P., Carvalho, F.F.R., 2021. Intake, digestibility, ingestive behavior and performance of goats fed spineless cactus genotypes resistant to carmine cochineal. Arq. Bras. Med. Vet. Zootec. 73 (5), 1209–1216. https://doi.org/10.1590/1678-4162-12381.

Niazifar, M., Danesh Mesgaran, M., Vakili, A.R., Eghbalsaied, S., Alimirzaei, M., Jafari, A., Foroughi, A.R., 2024. Slow-release non-protein nitrogen sources in animal nutrition: A review. Heliyon 10, e30543. https://doi.org/10.1016/j.heliyon.2024.e30543.

Nichols, K., Rauch, R., Lippens, L., Seymour, D.J., Martín-Tereso, J., 2023. Dose response to post-ruminal urea in lactating dairy cattle. J. Dairy Sci. 106, 23402. https://doi.org/10.3168/jds.2023-23402.

Pastorelli, G., Serra, V., Vannuccini, C., Attard, E., 2022. Opuntia spp. as alternative fodder for sustainable livestock production. Animals 12, 1597. https://doi.org/10.3390/ani12131597.

Ramalho, I.O., Santos-Souza, W., Homem, B.G.C., Costa-Soares, S., Borré, J.G., Ramos, J. V.B., Salgado, L., 2024. Nitrogen metabolism in ruminants and the implications of excess dietary nitrogen in pasture-based livestock farming. Zootec. Tóp. Atuais Pesq. Cient. Digit, 5, 62–80. https://doi.org/10.37885/240316172.

Rufino, L.M.A., Detmann, E., Gomes, D.Í., Reis, W.L.S., Batista, E.D., Valadares Filho, S. C., Paulino, M.F., 2016. Intake, digestibility and nitrogen utilization in cattle fed tropical forage and supplemented with protein in the rumen, abomasum, or both. J. Anim. Sci. Biotechnol. 7, 11. https://doi.org/10.1186/s40104-016-0069-9.

Sampaio, C.B., Detmann, E., Lazzarini, I., Souza, M.A., Paulino, M.F., Valadares Filho, S. C., 2009. Rumen dynamics of neutral detergent fiber in cattle fed low-quality tropical forage and supplemented with nitrogenous compounds. Rev. Bras. Zootec. 38, 560–569. https://doi.org/10.1590/S1516-35982009000300023.

Sánchez, B.M.S., Véras, A.S.C., Freitas, E.V., Farias, L.R., Albuquerque, J.G.S.S., Almeida, G.A.P., Mora-Luna, R.E., Monteiro, C.C.F., Gama, M.A.S., Ferreira, M.A., 2022. Partial replacement of sugarcane with cactus (Opuntia stricta) cladodes improves milk yield and composition in Holstein dairy cows. Anim. Prod. Sci. 62, 691–699. https://doi.org/10.1071/AN19648.

Santos, D.C., Batista, Â.M.V., Véras, A.S.C., Siqueira, M.C.B., Oliveira, J.S., Siqueira, M. C., Maciel, M.V., Carvalho, F.F.R., 2022. Sugarcane replaced by cactus cladodes improves the ruminal dynamics of sheep. Small Rumin. Res 213, 106715. https://doi.org/10.1016/j.smallrumres.2022.106715.

Santos, S.A., Carvalho, G.G.P., Azevêdo, J.A.G., Zanetti, D., Santos, E.M., Pereira, M.L.A., Mariz, L.D.S., 2021. Metabolizable protein: 1. Predicting equations to estimate microbial crude protein synthesis in small ruminants. Front. Vet. Sci. 8, 650248. https://doi.org/10.3389/fvets.2021.650248.

Shi, R., Dong, S., Mao, J., Wang, J., Cao, Z., Wang, Y., Li, S., Zhao, G., 2023. Dietary neutral detergent fiber levels impacting dairy cows' feeding behavior, rumen fermentation, and production performance during the period of peak lactation. Animals 13, 2876. https://doi.org/10.3390/ani13182876.

Silva, A.E.M., Franco, A.M., Solomon, J.K.Q., da Freiria, L.B., de Moura, F.H., Mazza, P. H.S., Birkenstock, B., Bezerra, L.R., Shenkoru, T., Fonseca, M.A., 2025. Cactus (Opuntia ficus-indica) diets reduce voluntary water intake, water footprint and enteric methane production improving ruminal fermentation in steers. J. Arid Environ. 227, 105311. https://doi.org/10.1016/j.jaridenv.2024.105311.

Silva, A.E.M., Lira, A.T., Ferreira, M.A., Barros, L.J.A., Melo, T.T.B., Siqueira, T.Q., Costa, C.T.F., 2015. Bagaço de cana-de-açúcar como volumoso exclusivo em dietas para ovinos. Rev. Bras. Saude Prod. Anim. 16, 118–129. https://doi.org/10.1590/S1519-99402015000100013.

Siqueira, M.C., Chagas, J.C., Monnerat, J.P.I., Monteiro, C.C., Mora-Luna, R.E., Felix, S. B., Ferreira, M.A., 2022. Cactus cladodes (Opuntia or Nopalea) and by-products of low nutritional value as solutions to forage shortages in semiarid areas. Animals 12, 3182. https://doi.org/10.3390/ani12223182.

Siqueira, T.D.Q., Monnerat, J.P.I., Chagas, J.C., Conceição, M.G., Siqueira, M.C.B., Viana, T.B.L., Ferreira, M.A., 2019. Cactus cladodes associated with urea and sugarcane bagasse: an alternative to conserved feed in semi-arid regions. Trop. Anim. Health Prod. 51, 1975–1980. https://doi.org/10.1007/s11250-019-01895-1.

Soares, M.S., Pires, A.J.V., Silva, L.G., Guimarães, J.O., Machado, T.C., Frazão, O.S., 2015. Utilização do bagaço de cana-de-açúcar na alimentação de ruminantes. Rev. Eletr. ônica Nutr. 12, 3837–3855.

Souza, N.K.P., Detmann, E., Valadares Filho, S.C., Costa, V.A.C., Pina, D.S., Gomes, D.I., Queiroz, A.C., Mantovani, H.C., 2013. Accuracy of the estimates of ammonia concentration in rumen fluid using different analytical methods. Arq. Bras. Med. Vet. Zootec. 65, 1752–1758. https://doi.org/10.1590/S0102-09352013000600021.

Van Soest, P.J., 1994. Nutritional ecology of the ruminant, 2nd ed. Cornell University Press, Ithaca, NY, USA.

Van Soest, P.J., Robertson, J.B., Lewis, B.A., 1991. Methods for dietary fiber, neutral detergent fiber, and nonstarch polysaccharides in relation to animal nutrition. J. Dairy Sci. 74, 3583–3597. https://doi.org/10.3168/jds.S0022-0302(91)78551-2.

Weiss, W.P., Conrad, H.R., St. Pierre, N.R., 1992. A theoretically based model for predicting total digestible nutrient values of forages and concentrates. Anim. Feed Sci. Technol. 39, 95–110. https://doi.org/10.1016/0377-8401(92)90034-4.

Zhang, J., Zheng, N., Shen, W., Zhao, S., Wang, J., 2020. Synchrony degree of dietary energy and nitrogen release influences microbial community, fermentation, and protein synthesis in a rumen simulation system. Microorganisms 8, 231. https://doi.org/10.3390/microorganisms8020231.

Zhang, T., Zhao, X., Li, Y., Liu, H., Xu, G., Chen, L., 2025. Effects of dietary fiber levels on ruminal fermentation characteristics, microbial community and nutrient digestibility in sheep. Small Rumin. Res. 238, 107289. https://doi.org/10.1016/j.smallrumres.2024.107289.